

Muhammad Saram Hassan

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EDUCATION

Lahore University of Management Sciences (LUMS)

Aug 2022 - Present

Bachelor of Science in Computer Science

- **Dean's Honor List (2022-2025)**
- **Relevant Coursework:** Advanced Topics in ML[†], Foundations of Generative AI[†], Deep Learning, Human Computer Interaction, Topics in Network Security[†], Distributed Systems[†], Network-centric Computing[†] [†]*Graduate-level courses*

PUBLICATIONS

Data Doping or True Intelligence? Evaluating the Transferability of Injected Knowledge in LLMs

arXiv:2505.17140 | **EMNLP Findings 2025** | Acceptance Rate: **17.35%**

Essa Jan, Moiz Ali, Muhammad Saram Hassan, Fareed Zaffar, Yasir Zaki.

RESEARCH EXPERIENCE

Research Assistant - TPI Lab, LUMS & CommNets Lab, NYUAD

Jan 2025 – Jul 2025

Supervised by: Dr. Fareed Zaffar (LUMS) and Dr. Yasir Zaki (New York University Abu Dhabi)

Evaluating the Transferability of Injected Knowledge Across Model Scales and Architectures

- **Research Question:** How does the task type (e.g., token-to-token mapping vs. content understanding) and model size influence the knowledge retention, integration, and generalizability during fine-tuning?
- **Approach:** Evaluated 5 model families (across 1.5B-72B parameters) on surface vs. comprehension tasks, comparing retention and scaling trends.
- **Findings:** Discovered that comprehension-based tasks (e.g. question answering) achieved 48% retention versus 17% for surface-level mapping tasks (e.g translation), revealing that deeper semantic integration enhances transferability.
- Observed consistent scaling improvements in knowledge retention from 38% to 72% across models for all tasks, revealing systematic advantages of scale for conceptual knowledge integration.
- *Published at EMNLP Findings 2025*

Research Intern - Stanford Research Institute (SRI) International

Aug 2025 – Present

Supervised by: Dr. Ashish Gehani and Dr. Ramneet Kaur

Illusive Understanding: Quantifying True Conceptual Understanding in LLMs

- **Research Question:** Do language models demonstrate true conceptual understanding that is robust across tasks and contexts, or do they rely on spurious correlations that break down under distribution shifts?
- **Approach:** Developing model-agnostic evaluation combining behavioral consistency analysis and representational geometry to assess how concept coherence is encoded and maintained across different layers of transformer models
- **Key Contributions:** Proposed dual-metric system: Behavioral Consistency Score and the Representation Consistency Score to quantify conceptual coherence, introduced Concept Coherence Index (CCI) to evaluate conceptual understanding across tasks and extend the framework with the Latent Direction Score (LDS), which enables causal evaluation of mechanistic interventions to probe concept understanding.

Research Assistant - Security & Privacy Lab, LUMS

Jul 2025 – Present

Supervised by: Dr. Fareed Zaffar (LUMS) and Dr. Taha Khan (Carnegie Mellon University)

Robustness of Web Agents in the Wild: Adversarial Influence on Autonomous Agents

- **Research Question:** How vulnerable are autonomous LLM-based web agents to adversarial manipulation by malicious websites, and what behavioral signatures enable detection of agent traffic?
- **Approach:** Combined large-scale web crawling and controlled adversarial environment design to evaluate robustness against hidden prompt injection, DOM manipulation, and visual spoofing.
- **Contributions:** Designed adversarial HTML testbed; developed behavioral signature analysis framework (HTTP headers, DOM traversal, timing); analyzing agentic footprints including cookie persistence, ad-tracking, and robots.txt/CAPTCHA compliance.
- **Impact:** Addresses emerging threat landscape of LLM-based agents interacting with adversarial websites, informing adaptive defenses.

*Supervised by: Dr. Muhammad Tahir (Associate Professor, LUMS)***Bias-Aware Interpretable Pruning for Robust Vision Models**

- **Research Question:** How does neural network compression affect inductive biases (texture, shape, color), and can we selectively preserve or remove biases to improve OOD robustness?
- **Approach:** Developed bias-aware pruning framework combining property-amplified augmentation with gradient-saliency and magnitude-based importance estimation for CNNs and ViTs.
- **Contributions:** Implemented interpretable pruning methodology for selective bias retention/removal; introduced filter-locking mechanism preserving domain-invariant features.

PROJECTS

Voice AI for Regional Languages | Uplift AI Partnership (YC startup) Jun 2024 - Jul 2024

- Led a cross-functional team of **100+ students** with Uplift AI to collect and curate **35,000+** regional language speech samples, directly contributing to the startup securing funding from Y-Combinator.
- Designed quality control and preprocessing pipelines ensuring **95%+** transcription accuracy, directly contributing to UpliftAI's successful seed funding round from Y-Combinator.

Model Compression via Structured and Unstructured Pruning | PyTorch Aug 2024

- Applied L2-norm unstructured and channel-wise structured pruning to VGG16, achieving **65.9%** post-fine-tuning accuracy (**7.5% lift**) from the baseline.
- Implemented PTQ and QAT across multiple bit-widths (FP16, BF16, INT8, INT4), visualizing accuracy-precision scaling trends.

Evaluating Architectural Bias and Generalization in Vision Models | PyTorch Mar 2024

- Analyzed robustness of ResNet, ViT, and CLIP under covariate and semantic shifts using controlled OOD augmentations.
- Assessed architectural bias through feature alignment and degradation under texture and edge transformations.

Knowledge Distillation Techniques via Feature Alignment and KL Divergence | PyTorch, GradCAM

- Compared Logit Matching, Label Smoothing, and Decoupled KD methods under domain shifts and adversarial attacks using GradCAM visualizations.

Federated Learning Under Data Heterogeneity | Go, Raft Protocol Oct 2024 - Dec 2024

- Benchmarked FL algorithms under Dirichlet-controlled non-IID settings, showing severe skew causes strong client drift; theoretically confirmed FedSGD matches centralized learning under full-batch IID.
- Evaluated SCAFFOLD, FedGH, and FedSAM: Found SCAFFOLD to be most robust under extreme heterogeneity, FedGH effective in moderate skew, and FedSAM improving generalization via flatter minima.

Raft-Based Fault-Tolerant Distributed Key-Value Store | Go, Raft Protocol Oct 2024 - Dec 2024

- Implemented full Raft consensus in Go ensuring consistency via leader election, log replication, and state reconciliation.
- Built a distributed KV-store with strong consistency and crash recovery mechanisms.

TEACHING EXPERIENCE

CS-210: Discrete Mathematics | Dr. Malik Jahan, LUMS**CS-100: Computational Problem Solving | Dr. Fareed Zaffar, LUMS****Fundamentals of Machine Learning | LUMS Science School & UpliftAI**

- Designed and taught curriculum covering supervised/unsupervised learning, neural networks, and practical ML applications.

Exploring the Universe with Code: CS for Astronomy | FAST University

- Designed and taught interdisciplinary course combining computational thinking with astronomy for 100+ students.

TECHNICAL SKILLS

Languages: Python, Haskell, Go, C/C++, JavaScript, C#**Frameworks:** React, Node.js, WordPress, PyTorch, Hugging Face (Transformers, Diffusers, timm, unsloth)**Tools:** Git, Docker, VS Code, LLVM, LangChain,